

Sample-phase spatial attention in the noisy-memory Luo task

Noise-averaged maps, oracle spatial inhibition, psychometrics, chronometrics, and one complete rendered trial

Frozen checkpoint	Iteration 12899; SHA-256 2c4a6d2d21bd9698387f7a4b27cbc242c7f3bb097868518deb7790ae1c8b339f
Primary intervention	Bias -6.0 at all visual and memory keys in the future-tested quadrant, sample frames t0–t1 only.
Primary result	Suppressing the future-tested sample region produced 0 discordant correctness outcomes among n=704 paired trials (observed Δ accuracy +0.000). The ordinary paired bootstrap is degenerate because every observed pair agrees; the conservative two-sided 95% no-discordance bound is $ \Delta \text{ accuracy} < 0.0052$ under IID trial-pair sampling.
Run	luo2015_scientific_assay_20260803_production

Abstract

We evaluated one frozen recurrent vision policy in a seven-frame Luo–Maunsell-inspired change-detection task. Attention maps average 64 repeats of each exactly fixed latent condition while sensory, mnemonic, and policy-sampling noise vary. The causal assay suppressed routing to the future-tested sample quadrant and used equal-area active-sample and blank-region controls. Suppressing the future-tested sample region produced 0 discordant correctness outcomes among $n=704$ paired trials (observed Δ accuracy $+0.000$). The ordinary paired bootstrap is degenerate because every observed pair agrees; the conservative two-sided 95% no-discordance bound is $|\Delta \text{ accuracy}| < 0.0052$ under IID trial-pair sampling. The manipulation reduced sample-frame mass at the future-tested region from 0.2342 to 0.0011. All estimates come from one mid-training checkpoint and do not quantify training-seed uncertainty.

Question and evidence boundary

The task has no symbolic or spatial cue. Two sample Gabors appear simultaneously at locations 0 and 3. The lesion uses oracle knowledge of which sample location will later be tested; the model does not receive that information.

Evidence boundary. Raw attention maps are routing weights, not orientation-content decoding. A regional attention lesion tests the effect of changing that routing computation, but does not identify the content stored in recurrent tokens.

Frozen model and protocol

The checkpoint uses a 20×20 token grid (400 current-image queries; 400 visual plus 400 recurrent-memory keys), sensory orientation-noise SD 5.0° , and mnemonic-noise SD 0.64. Training sampled Δ uniformly from -35° to $+35^\circ$. Actions were sampled. Each exact map cell used 64 noisy repetitions. Fixed-magnitude behavioral curves used 64 changed trials per point and 128 common no-change trials.

Behavioral effect of sample inhibition

Condition	Correct	Accuracy	Sample mass	Paired Δ ; bound	HR 18°	d' 18°
Natural	476/704	0.676	0.2342	reference	0.984	3.443
Inhibit future-tested sample	476/704	0.676	0.0011	$+0.000$; $ \Delta < 0.0052$	0.984	3.443
Inhibit other active sample	476/704	0.676	0.0012	$+0.000$; $ \Delta < 0.0052$	0.984	3.443
Inhibit blank control	476/704	0.676	0.0009	$+0.000$; $ \Delta < 0.0052$	0.984	3.443

Suppressing the future-tested sample region produced 0 discordant correctness outcomes among $n=704$ paired trials (observed Δ accuracy $+0.000$). The ordinary paired bootstrap is degenerate because every observed pair agrees; the conservative two-sided 95% no-discordance bound is $|\Delta \text{ accuracy}| < 0.0052$ under IID trial-pair sampling.

Phase-resolved routing

At first-test onset t_3 , combined incoming attention favored the other active-sample quadrant over the tested quadrant: tested-minus-other mass was -0.0615 at L_0 (95% interval $[-0.0615, -0.0615]$) and -0.0608 at L_3 . Among the small survivor sets reaching t_4 on changed trials, recurrent-memory routing then favored the tested quadrant: $+0.0331$ at L_0 ($n=8$) and $+0.0256$ at L_3 ($n=11$). At guaranteed test 2, the memory contrast reversed by location ($+0.0519$ at L_0 versus -0.0517 at L_3), so it does not support a location-general same-site return.

Psychometric and chronometric shape

The controlled psychometric slice was strongly non-monotonic: natural hit rate and d' peaked at 18° (0.984, 3.443) but fell to 0.547 and 1.565 at 35° . Thus a monotone sigmoid would be a poor summary of this checkpoint. All intervention hit-rate and d' points were identical. The only visible timing deviation was a -0.018 -frame mean-hit shift at 12° under tested-sample inhibition, without a correctness change.

Sample-phase lesion and manipulation check

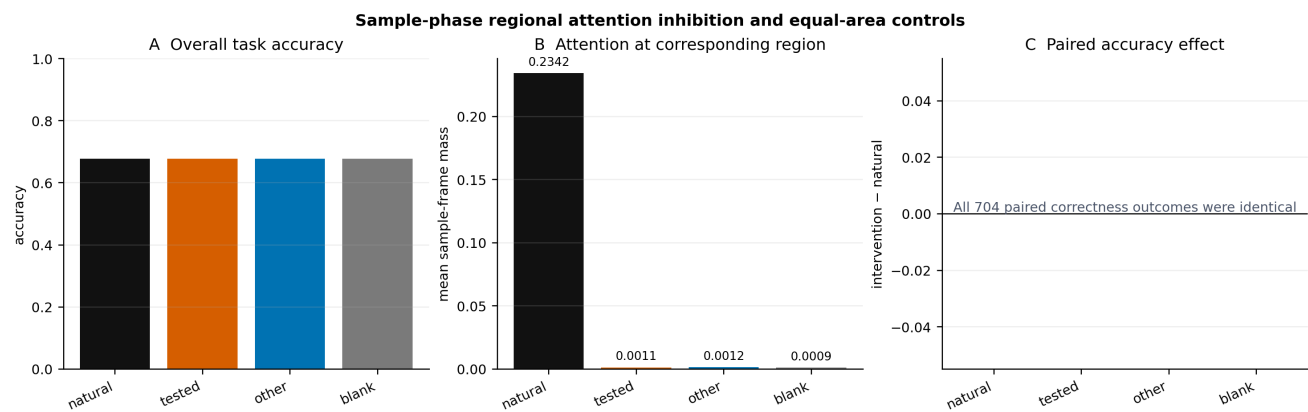


Figure 1. Sample-phase lesion and manipulation check. Accuracy pools all controlled changed trials and the common no-change bank.

Psychometric and chronometric slices

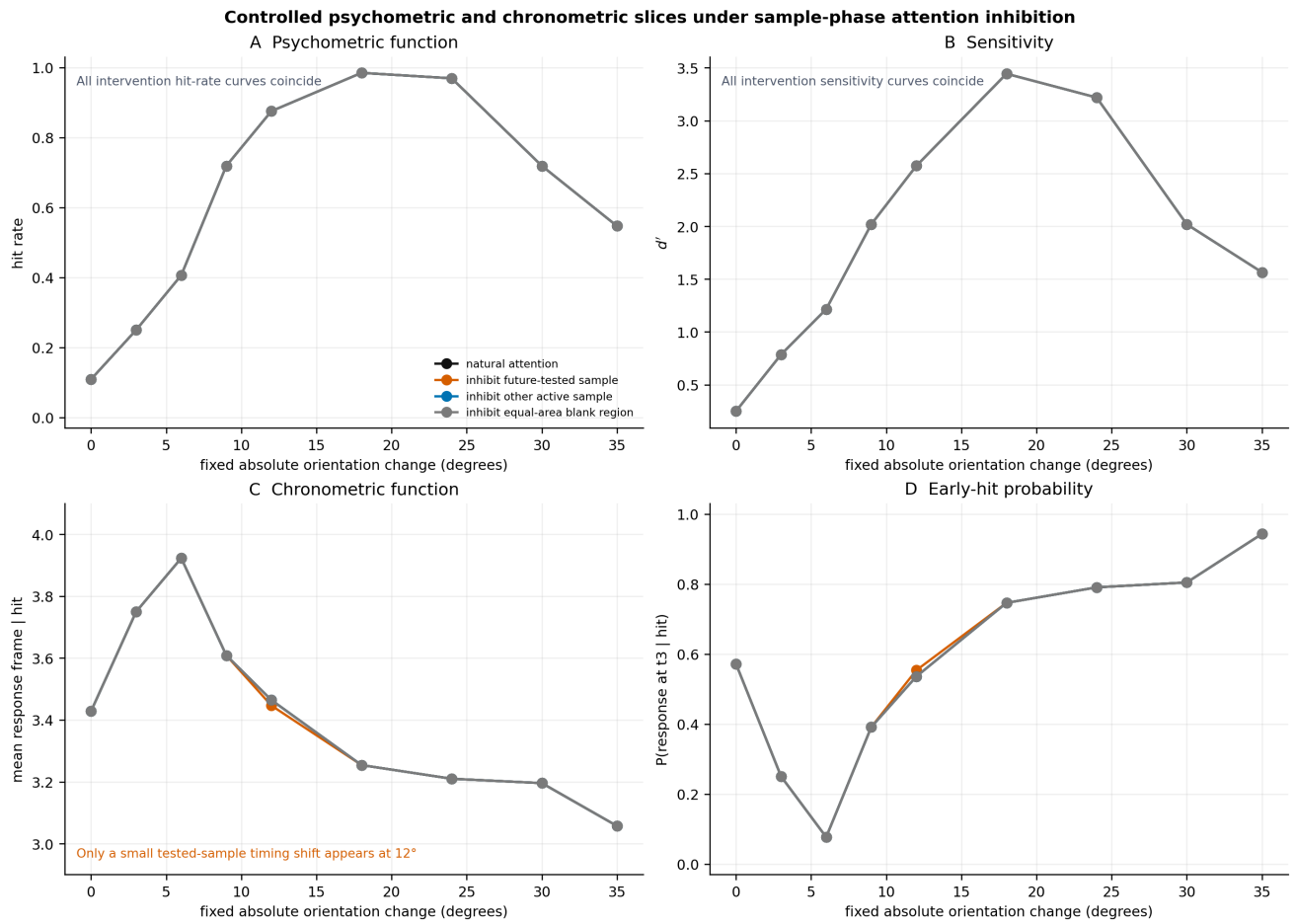


Figure 2. Psychometric and chronometric slices. Response time is a logical frame index, not biological milliseconds.

Combined visual-plus-memory incoming attention across four exact latent conditions and all seven logical frames

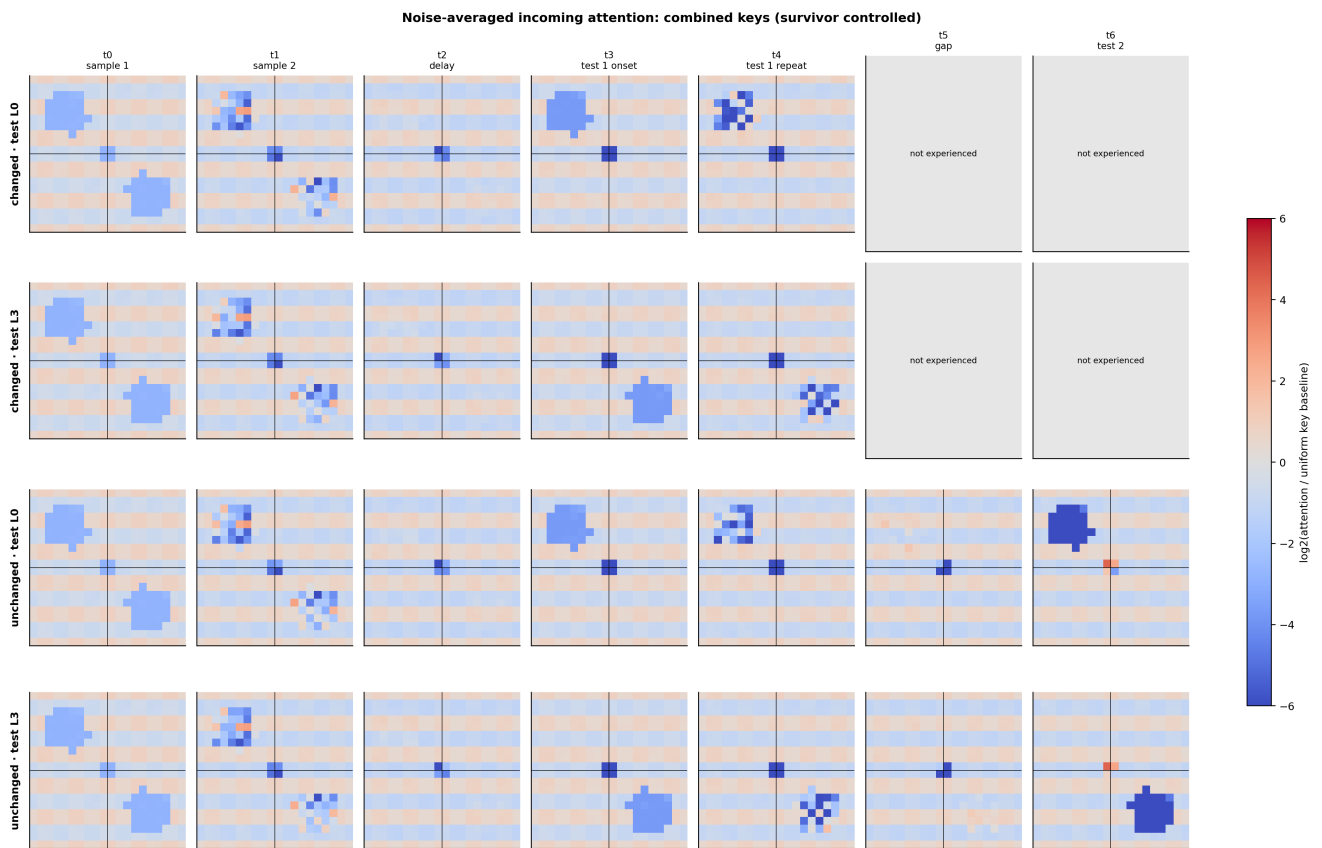


Figure 3. Combined visual-plus-memory incoming attention across four exact latent conditions and all seven logical frames. Gray frames were not experienced after termination.

Recurrent-memory source maps under the joint softmax

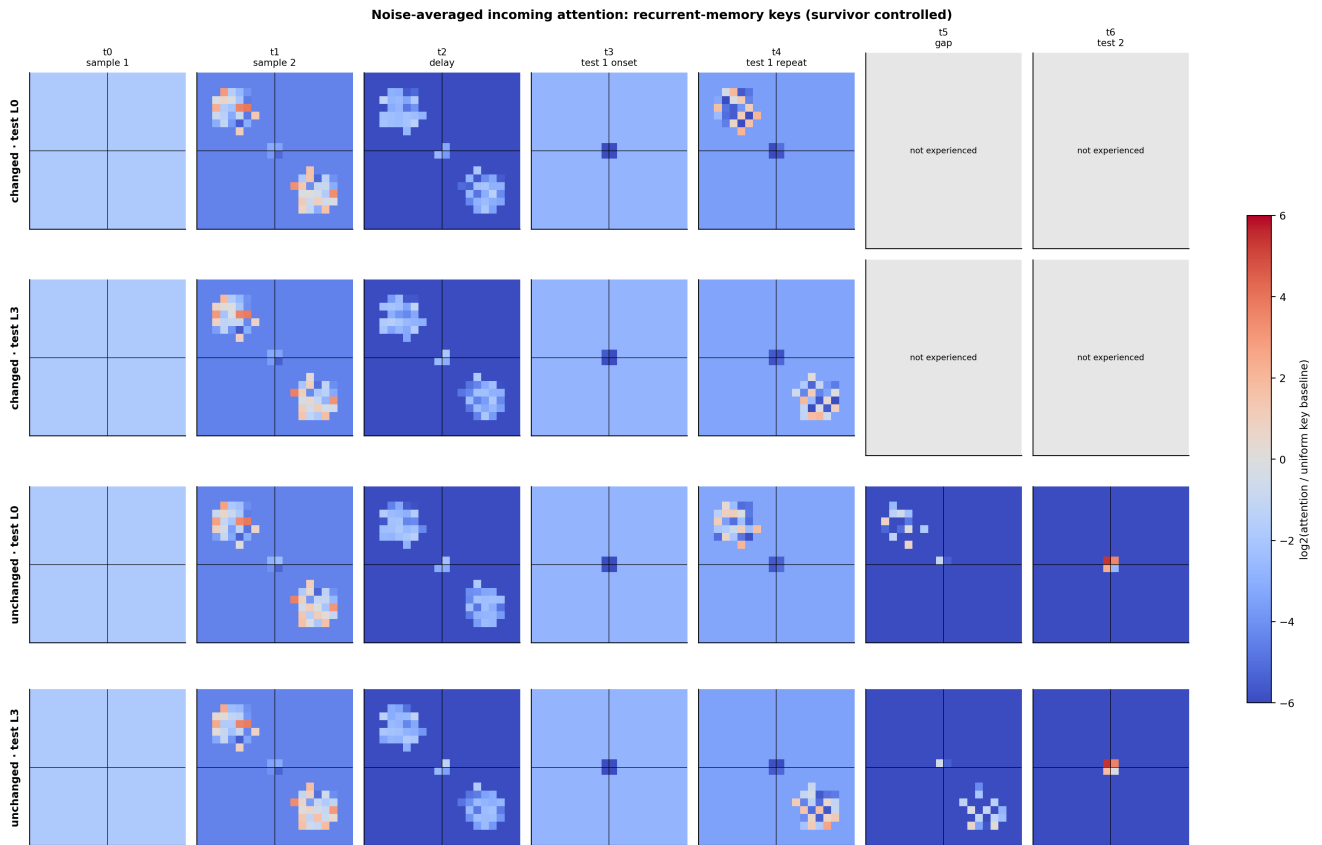


Figure 4. Recurrent-memory source maps under the joint softmax. Sources are not independently renormalized.

Current-visual source maps under the same fixed conditions, noise repetitions, and survivor rule

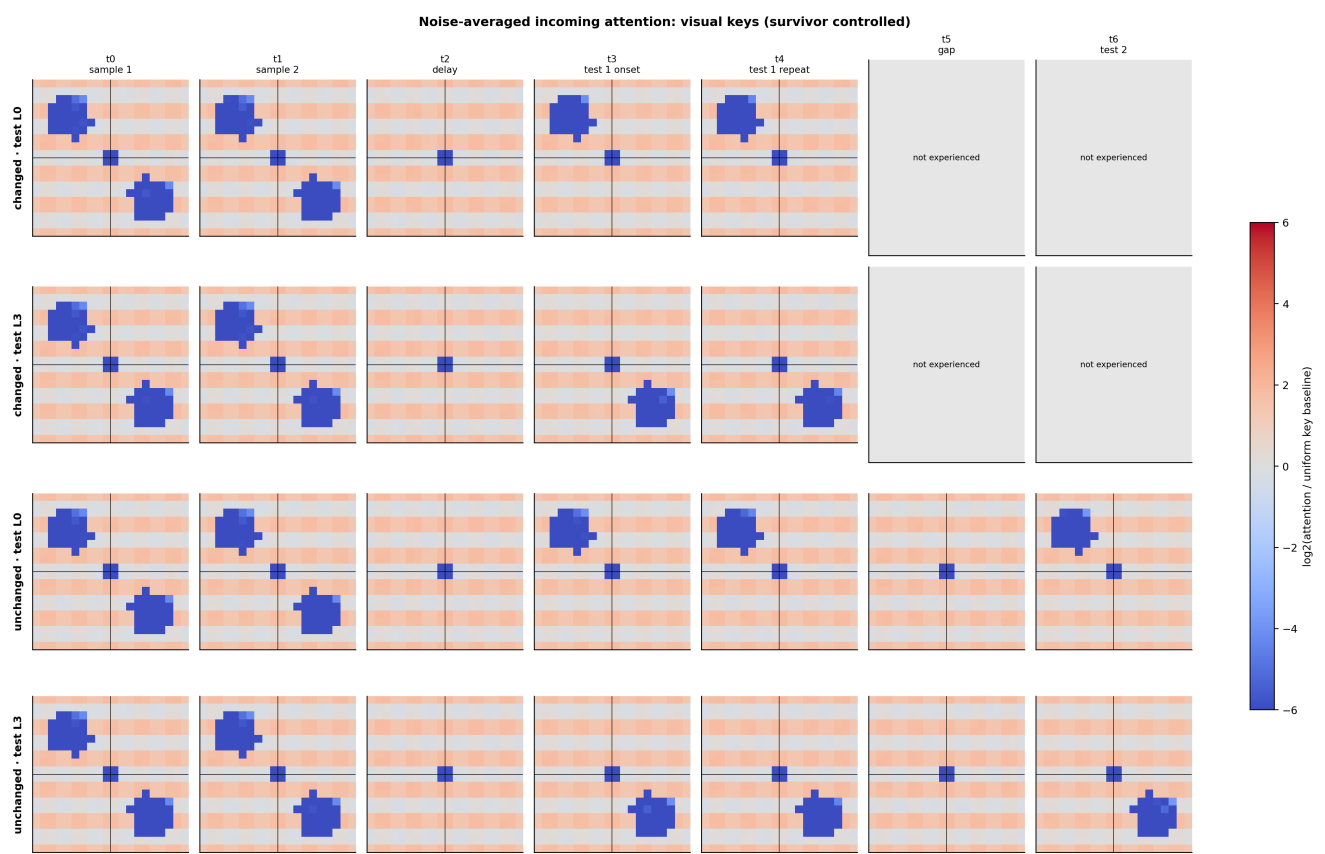


Figure 5. Current-visual source maps under the same fixed conditions, noise repetitions, and survivor rule.

Query-quadrant to key-quadrant routing

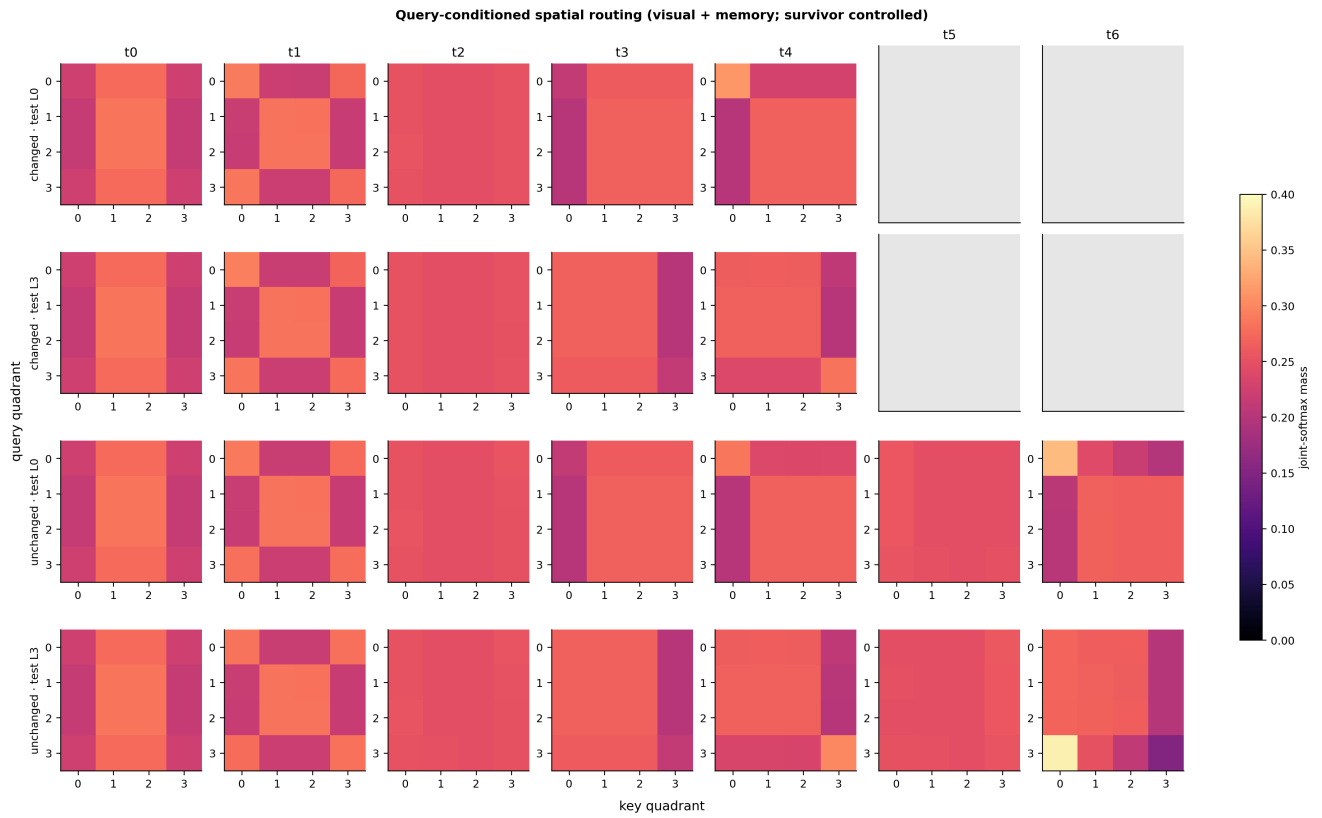


Figure 6. Query-quadrant to key-quadrant routing. Each 4×4 tile preserves query origin and spatial destination.

One complete trial

The example had an unchanged first test at location 0 followed by a guaranteed $+18^\circ$ second test. The sampled policy first declared at t_6 , producing `correct_rejection`. Declare probabilities were $[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0]$; actions were $[0, 0, 0, 0, 0, 0, 1]$ (0=wait, 1=declare).

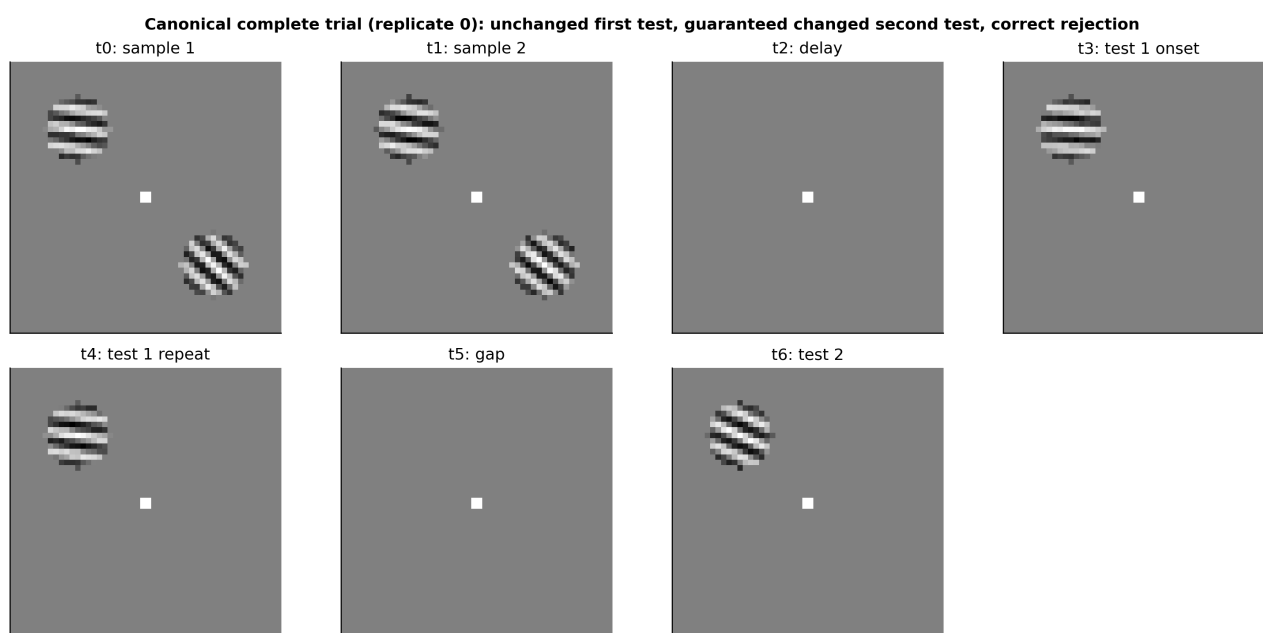


Figure 7. Every physical input frame in the specific trial. The companion GIF pairs each image with incoming attention and action probability.

Interpretation and limitations

(1) This is one frozen checkpoint; intervals cover evaluation-trial noise, not independent training seeds. (2) The finite -6 logit bias is strong soft suppression, not hard exclusion: 0.48% of the natural target-region mass remained. It suppresses both visual and memory keys in a quadrant under a joint softmax, changing competitive routing rather than deleting pixels. (3) Oracle targeting is a mechanistic counterfactual, not an endogenous cue. (4) Fixed-magnitude curves condition the training sampler and must not be called same-distribution performance. (5) The biological target is Luo and Maunsell (2015) V4; cross-attention is a computational abstraction, not a firing-rate measure.

Reproducibility

Checkpoint SHA-256: 2c4a6d2d21bd9698387f7a4b27cbc242c7f3bb097868518deb7790ae1c8b339f. The report directory retains source-resolved reduced attention arrays, query-routing arrays, CSV curves, every trial frame, the GIF, producer hashes, and an artifact manifest. The mutable live-training checkpoint was never opened.

Reference

Luo, T. Z. & Maunsell, J. H. R. (2015). Neuronal modulations in visual cortex are associated with only one of multiple components of attention. *Neuron* 86, 1182–1188. doi:10.1016/j.neuron.2015.05.007.